

Heart Rate Estimation from Noisy PPGs Using 1D/2D Conversion and Transfer Learning

Emil Dark
VTT Technical Research Centre of
Finland Ltd.
Oulu, Finland
emil.dark@vtt.fi

Umer Saleem
VTT Technical Research Centre of
Finland Ltd.
Oulu, Finland
umer.saleem@vtt.fi

Arttu Lämsä
VTT Technical Research Centre of
Finland Ltd.
Oulu, Finland
arttu.lamsa@vtt.fi

Constantino Álvarez Casado
University of Oulu.
Oulu, Finland
constantino.alvarezcasado@oulu.fi

Miguel Bordallo López
University of Oulu & VTT Technical
Research Centre of Finland Ltd.
Oulu, Finland
miguel.bordallo@vtt.fi

ABSTRACT

In this article, we introduce a new approach for estimating the heart rate from noisy photoplethysmography (PPG) signals. We propose the use of two-dimensional representations of signals that are fed into a residual deep neural network that performs the regression task. Our approach leverages transfer learning and pre-trained models to further reduce the prediction error, resulting in state-of-the-art results in a challenging benchmark dataset.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks; Supervised learning by regression; Transfer learning; Mathematics of computing** → *Time series analysis*.

KEYWORDS

Deep learning, Heart rate estimation, PPG

ACM Reference Format:

Emil Dark, Umer Saleem, Arttu Lämsä, Constantino Álvarez Casado, and Miguel Bordallo López. 2022. Heart Rate Estimation from Noisy PPGs Using 1D/2D Conversion and Transfer Learning. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544793.3563407>

1 INTRODUCTION

Extracting vital signs in a non-invasive manner is a topic that has been widely studied in the academic field. Wearable devices can be used to measure photoplethysmography (PPG) signals that represent changes in blood volume in the time domain. Vital signs, such as heart or breathing rate, provide important information and enable a person to make decisions or take required actions. PPGs

and vital signs derived from them have numerous applications, including the assessment of sleep cycles[6], stress[19], mental health problems[5], or even self-induced states such as exercise[12] or meditation[1]

To extract this information from the PPG signal, a traditional approach is to use simple signal processing such as peak counting or frequency estimation. However, the raw PPG signals obtained from wearable devices are often noisy and subject to several motion artifacts. Obtaining reliable parameters such as heart-rate is hence a very challenging task. In this context, several machine learning approaches have been explored in the literature. In this work, we introduce a deep neural network based approach that utilizes models pre-trained with common image datasets and uses transfer learning to regress heart rate from a PPG signal. Our contributions are detailed as follows:

- We convert 1D PPG images into 2D representations by using column-normalized spectrograms, and use them as an input for 2D residual networks (ResNet50) to perform regression of the heart rate from noisy signals.
- We explore multiple training configurations, and evaluate the impact of transfer learning from pretrained weights (ImageNet) in the accuracy of the models.
- We perform experiments in a challenging dataset (PPG-DaLiA) composed by people performing sport activities. We obtain state-of-the-art results for PPG-only approaches that do not use additional sensor data.

2 RELATED WORK

Heart rate estimation from PPG signals has been thoroughly investigated in the literature[13]. Studies that attempt to predict heart rate from PPG signals can broadly be classified into signal processing-based methods and machine learning-based methods. Signal processing techniques exploit the characteristics of the signal for heart rate estimation. Typical approaches include peak counting or frequency estimation[23][4].

Supervised machine learning-based techniques employ a set of labeled training samples to learn statistical models. Once trained, these models can subsequently be employed to estimate the heart



This work is licensed under a Creative Commons Attribution International 4.0 License.

UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom
© 2022 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9423-9/22/09.
<https://doi.org/10.1145/3544793.3563407>

rate from unseen signals[22]. Several attempts to create these models from one-dimensional signals are relatively straightforward. For example, Shahid Ismail et al. [10] proposed to extract a set of temporal, spectral, and time-frequency features that characterize the PPG signal. These features fed a hybrid convolutional-recurrent neural network (C-RNN) in a regression framework, composed by 1D convolutions with long short-term memory layers. The method has been shown to be effective in estimating the heart rate from PPG signals acquired from people performing a variety of exercises.

Modern deep learning approaches have inspired new image classification methods by using different architectures of convolutional neural networks designed to exploit invariances and capture compositional features in the data, harnessing the benefits of a very deep network [14], [7], [8]. These systems have been largely developed to deal with image data. Transforming time-series data to 2-dimensional representations such as spectral representations has proven to be a successful strategy, effectively framing the problem as a visual task. This strategy has been used in e.g. EEG data [17], and using them as input to different neural networks has been relatively successful in similar tasks. For example, Vilamala et al. [24] used VGGNet16 (a very deep convolutional network containing 16 layers) with their EEG spectrograms for several classification tasks.

3 DATA AND METHODS

3.1 Dataset

In our experiments, we employ a challenging dataset for heart-rate estimation from noisy PPG data, PPG-DaLiA [15] dataset.

The dataset contains several signals recorded from a wrist-worn (Empatica E4) and a chest-worn (RespiBAN) device. In our experiments we used as an input only the PPG signal recorded from the wrist, using Empatica E4 device. The PPG signal is sampled at 64 Hz and contains one channel. As a ground truth, we used the heart rate values provided on the dataset, that were derived from the accurate chest-worn ECG device.

The dataset is composed of data recorded from 15 subjects (8 women and 7 men, aged 21 to 55 years) during activities of daily life, including sports, and contains numerous motion artifacts. In total, the dataset contains more than 36 hours of data. In the PPG-DaLiA dataset, most of the subjects have approximately the same amount of data measured, with only subject 6 being an outlier with comparatively less data. Based on the PPG-DaLiA dataset documentation [15], the heart rate of subject 5 was mentioned to be an obvious outlier compared to the other subjects and we expect to see this observation in our results too.

3.2 Methods

The approach described in this paper uses windows extracted solely from PPG signals, which are transformed into spectrograms for a two-dimensional representation that is directly interpretable by visual inspection. The heart rate for each signal window is then regressed by feeding these spectrograms into models based on residual 2D convolutional neural networks.

3.2.1 Preprocessing. Before the signal is used as input for the 2D CNN model, certain pre-processing steps are needed. The DaLiA dataset contains annotations for 8 second windows with 6 second

overlap between sequential windows. Hence, the raw PPG signals in the data set are first divided into time windows to match the annotations. Then the signal is bandpass-filtered with cut-off frequencies of 0.75 Hz and 3 Hz. This represents the heart rate range from 45 BPM to 180 BPM. After this, the 1D signal is turned into a 2D spectrogram of size 257x257x3.

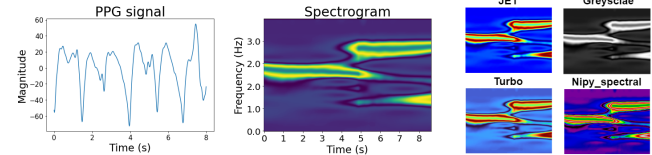


Figure 1: Example visualization of a PPG signal and corresponding spectrogram.

A spectrogram is a visual way to represent the strength or 'loudness' of a signal over time at various frequencies present in a particular waveform. We compose our spectrogram by computing the normalized power spectral density of consecutive overlapping samples and concatenating them column by column. Each one of the columns is normalized based on the maximum and minimum values. The full spectrogram is then cropped by excluding frequencies outside the [0.07Hz, 3.63Hz] range. This also removes frequencies from the spectrogram that are outside the range of normal human heart rate. With a careful selection of the number of spectral bins (256 bins in our desired spectral range) and the number of consecutive samples (256), the resulting spectrograms have a resolution that matches the default input size used in our approach. Spectrograms can be composed by using different color representations. We hypothesized that different colormaps represent varying amplitudes in a distinct way, a fact that could have an influence on model performance. In this context, we created the spectrogram using different colormaps, namely 'viridis', 'jet', 'grayscale', 'turbo', and 'nipy_spectral' as defined by matplotlib. Figure 1 shows an example visualization of the PPG signal and different spectrograms.

3.2.2 Model. After the data preprocessing, the spectrograms can be directly fed into a 2D neural network. For our experiments, we used the ResNet50 topology, adding a custom layer consisting of one neuron with linear activation function to transform a classification task into heart rate regression. The main idea of our approach is presented in Figure 2. The spectrograms obtained through the preprocessing steps are used as input for the ResNet50 part of the model. The output of this part is then fed directly to either a single neuron output layer or to a hidden layer connected to the output layer (depending on the experiment). In other words, the backbone of the ResNet50 is used for feature extraction from the spectrograms and then the regression for the heart rate is done based on these features.

3.2.3 Model tuning. Our method involves manually defining a subset of the hyperparametric space and exhausting all combinations of the specified hyperparameter subsets, also known as 'grid search'. The performance of each combination was then evaluated on the TensorBoard HPparams dashboard using the model histories. We

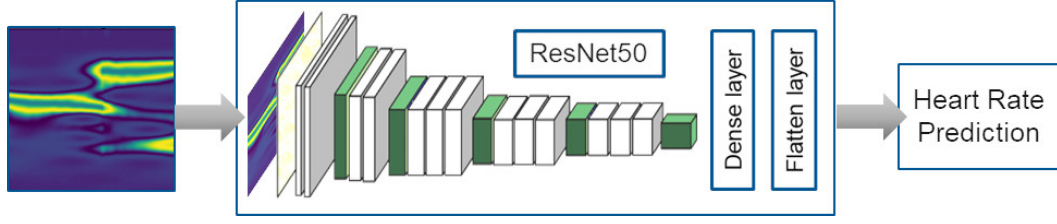


Figure 2: Neural network model "ResNet50 + linear output" used in the experiments.

selected optimal hyperparameters by using the loss value as a metric and visually inspecting the training curves. The search space was limited to the values shown in Table 1. Although our final models were computed using Leave-one-Subject-out validation, for the hyperparameter selection, we used a subset of the users as the testing set. The best results were obtained, using the default Adam optimizer with a smaller learning rate of $3e^{-4}$. The best loss function was the L1 distance between the predicted and ground truth values (mean average error). Although we tried training for up to 120 epochs, for most of the models, the best results were usually obtained around the 60 to 70 epoch mark.

Table 1: Hyperparameter search space, while tuning model

Parameter	Search space	Best value
Batch size	32, 64	64
Optimizer	'Adam', 'SGD', 'NAdam'	'Adam'
Loss	'L1', 'L2'	'MAE'
Learning rate	[0.001, 0.0001]	0.0003
Epochs	[25, 120]	120

4 EXPERIMENTS

The evaluation of the implemented approach was performed using leave-one-subject-out (LOSO) cross-validation with the DaLiA dataset. In this way, we obtained results for each one of the subjects, and we calculated the average error, the standard deviation of the error across subjects and the maximum and minimum error. We compare our results from a starting baseline model (model B0) that simply takes the maximum peak of the power spectral density computed using Welch's method, and we calculate the error against the ground-truth value.

We trained and tested on the whole dataset using only a subset of all the possible combinations of hyperparameters, following the strategy mentioned above. We divided our training into three different configurations. During the first configuration (models F1-3), we initialized ResNet50 using the weights of a pre-trained model using ImageNet obtained by importing it using the *Keras* framework. We then froze the weights of the ResNet50 part of the model and trained only on the additional layer(s): a 128 neuron hidden layer with ReLU activation preceding the output layer (only in case of model F3) and the single neuron output layer with linear activation (models F1-F3). During the second configuration (models R4-10), we initialized the network using random values, and then proceed to train the entire architecture, including all layers of the ResNet50 architecture and the additional output layer, that

produced the regression using a linear activation function. For the third configuration (models IN11-15), we repeated the same training scheme that used the whole model, completely unfrozen, but initialized the weights with the pre-trained model on ImageNet.

For all configurations, we used a set of different colormaps, following the suggestions laid by Kovessi[11]. In this context, we trained complete models in colormaps, 'jet', 'turbo', 'nipy_spectral', 'grayscale', and 'viridis' (matplotlib default colormap). Table 2 shows a summary of the experiments for different configurations, ordered from worse to best in terms of Mean Average Error (MAE).

5 ANALYSIS OF RESULTS

The experiments were divided into three groups; training the last layers using pretrained networks with frozen weights; training the entire models using random initialization; and training the entire model initialized with ImageNet weights.

From the results, it can be concluded that initializing the network with pre-trained weights results in a smaller error than when initialization is done with random values. However, this effect is only noticeable when training the entire network, since using the features obtained by the pre-trained weights in a straightforward manner (frozen configuration) shows a considerable worse performance. This suggests that low-level features learned from a roughly related pretext task, in this case object detection, are useful for heart rate detection from noisy spectrograms, while higher-level features are possibly less relevant. Most likely this is due to different nature of both training and pre-training data; ImageNet contains images of common objects (e.g. vehicles, animals) which contain high-level features that are different from the ones that can be seen in spectrograms, while lower-level features such as color edges and borders are still relevant.

Analyzing the results in detail, we can see that using the frozen ImageNet weights and training only a linear output layer, the error is relatively high, 10.41 BPM. Adding a trainable hidden 128 neuron layer with ReLU activation the error decreases to 8.88 BPM. Training all the weights of the ResNet50 network provided significantly lower errors. When ResNet50 was initialized with random values, training the whole model (including ResNet50 and linear output) resulted in models ranging from 8.68 BPM to 7.99 BPM. However, initializing the network with pre-trained ImageNet weights but still training all the weights of the ResNet50 network consistently resulted in the lowest errors, ranging from 7.77 BPM to 7.28 BPM. Using pre-training weights seems to be a valid strategy for all possible configurations.

For each particular configuration, it seems that, in general, higher batch sizes and a sufficient number of epochs result in slightly better performance. The impact of the different colormaps seems to

Table 2: Different variations in the model configuration used in the experiments.

ID	Pre-training	Learning rate	Batch size	Colormap	Output layers	Epochs	MAE
B0 Base	-	-	-	-	-	-	15,05
F1	ImageNet	0,001	64	Viridis	Linear output*	100	10,41
F2	ImageNet	0,001	64	Greyscale	Linear output*	100	9,06
F3	ImageNet	0,001	64	Viridis	H128ReLU + Linear output *	100	8,88
R4	Random	0,001	32	Turbo	Linear output	50	8,68
R5	Random	0,001	32	JET	Linear output	50	8,27
R6	Random	0,001	64	JET	Linear output	50	8,13
R7	Random	0,001	64	Nipy_spectral	Linear output	100	8,12
R8	Random	0,001	64	Viridis	Linear output	100	8,08
R9	Random	0,001	32	JET	Linear output	100	8,05
R10	Random	0,001	64	JET	Linear output	100	7,99
IN11	ImageNet	0,001	64	Turbo	Linear output	100	7,77
IN12	ImageNet	0,001	64	Greyscale	Linear output	100	7,67
IN13	ImageNet	0,001	64	Viridis	Linear output	100	7,57
IN14	ImageNet	0,001	64	JET	Linear output	100	7,55
IN15	ImageNet	0,0003	64	JET	Linear output	100	7,28

* Train only last layers, all ResNet50 weights frozen

be relatively limited, although the 'JET' scheme lightly outperforms the other configurations. Finally, as suggested in the related literature[2], a small learning rate could improve the results even further.

5.1 Comparison with state of the art

We compared the results of our method with other state-of-the-art methods for heart rate regression by calculating the mean average error for leave-one-subject-out validation, showing the standard deviation across users, as well as the maximum and minimum error.

Table 3: Comparison of the performance of various methods in terms of average error $\pm$ STD, in DaLia dataset

Method	AVG. error $\pm$ STD	[min,max]
Schaeck2017 [20]	20.45 $\pm$ 7.1	8.72 , 33.05
SpaMa [18]	15.56 $\pm$ 7.5	9.08 , 39.28
SpaMaPlus [16]	11.06 $\pm$ 4.8	6.17 , 24.06
DeepPPG [15]	7.65 $\pm$ 4.2	3.91 , 18.51
ResNet50 IN15 (Ours)	7.28$\pm$3.5	3.29 , 12.68

The comparison indicates that our methods obtain results comparable to state-of-the-art methods, suggesting the validity of our approach. Several older methods [20][18], appeared to be less accurate than simple baselines computed with frequency transformations and peak detection (e.g. baseline B0). However, more recent methods based on spectrograms and deep learning show reasonable accuracy (e.g. DeepPPG[15]). These results are comparable to ours, but they are obtained at the cost of larger models, more costly spectral transformations, and the incorporation of additional signals. It should be noted that other methods have been able to further improve accuracy (approximately 20%), but are not directly

comparable, since they either use additional acceleration signals (NAS-PPG[21], CurToss[25]), or propose other validation protocols that incorporate data from all users in the training set (TAPIR[9], Q-PPG[3]).

6 CONCLUSIONS

In this article, we have shown how to accurately estimate heart rates from noisy PPG signals by carefully transforming them into 2-dimensional representations that allow us to leverage transfer learning strategies and pre-trained networks. Our results show how initializing the network with pre-trained weights obtained with a related visual task (object detection) outperforms approaches where the whole network is trained from random initialization. Our results show that this straightforward approach can outperform simpler approaches based on statistical feature extraction approaches, obtaining comparable and even superior results to state-of-the-art ad hoc neural network architectures. Future work should explore in a systematic manner different 2-dimensional representations such as those based on, e.g., wavelets, and the impact of pre-training in different neural network architectures.

ACKNOWLEDGMENTS

This research has been supported by the Academy of Finland 6G Flagship program under Grant 346208 and PROFi5 HiDyn under Grant 32629, and the InSecTT project, which is funded under the European ECSEL Joint Undertaking (JU) program under grant agreement No 876038.

REFERENCES

- [1] Constantino Álvarez Casado, Petteri Paananen, Pekka Siirtola, Susanna Pirttikangas, and Miguel Bordallo López. 2021. Meditation Detection Using Sensors from Wearable Devices. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers*. 112–116.

- [2] Peter Bermant. 2021. BioCPPNet: automatic bioacoustic source separation with deep neural networks. *Scientific Reports* 11 (12 2021), 23502. <https://doi.org/10.1038/s41598-021-02790-2>
- [3] Alessio Burrello, Daniele Jahier Pagliari, Matteo Risso, Simone Benatti, Enrico Macii, Luca Benini, and Massimo Poncino. 2021. Q-PPG: Energy-Efficient PPG-based Heart Rate Monitoring on Wearable Devices. *IEEE Transactions on Biomedical Circuits and Systems* PP (10 2021), 1–1. <https://doi.org/10.1109/TBCAS.2021.3122017>
- [4] Constantino Alvarez Casado and Miguel Bordallo López. 2022. Face2PPG: An unsupervised pipeline for blood volume pulse extraction from faces. *arXiv preprint arXiv:2202.04101* (2022).
- [5] Constantino Alvarez Casado, Manuel Lage Cañellas, and Miguel Bordallo López. 2022. Depression Recognition using Remote Photoplethysmography from Facial Videos. *arXiv preprint arXiv:2206.04399* (2022).
- [6] Eduardo Gil, Martin Mendez, Jose Maria Vergara, Sergio Cerutti, Anna Maria Bianchi, and Pablo Laguna. 2008. Discrimination of sleep-apnea-related decreases in the amplitude fluctuations of PPG signal in children by HRV analysis. *IEEE transactions on biomedical engineering* 56, 4 (2008), 1005–1014.
- [7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 770–778. <https://doi.org/10.1109/CVPR.2016.90>
- [8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Identity Mappings in Deep Residual Networks. *CoRR abs/1603.05027* (2016). [arXiv:1603.05027](http://arxiv.org/abs/1603.05027) <http://arxiv.org/abs/1603.05027>
- [9] Nicholas Huang and Nandakumar Selvaraj. 2020. Robust PPG-based Ambulatory Heart Rate Tracking Algorithm, Vol. 2020. 5929–5934. <https://doi.org/10.1109/EMBC44109.2020.9175346>
- [10] Shahid Ismail, Imran Siddiqi, and Usman Akram. 2022. Heart rate estimation in PPG signals using Convolutional-Recurrent Regressor. *Computers in Biology and Medicine* 145 (2022), 105470. <https://doi.org/10.1016/j.combiomed.2022.105470>
- [11] Peter Kovési. 2015. Good Colour Maps: How to Design Them. *CoRR abs/1509.03700* (2015). [arXiv:1509.03700](http://arxiv.org/abs/1509.03700) <http://arxiv.org/abs/1509.03700>
- [12] Mohammod Abdul Motin, Chandan Kumar Karmakar, and Marimuthu Palaniswami. 2019. PPG derived heart rate estimation during intensive physical exercise. *IEEE access* 7 (2019), 56062–56069.
- [13] Sungkyu Park, Marios Constantinides, Luca Maria Aiello, Daniele Quercia, and Paul Van Gent. 2020. Wellbeat: A framework for tracking daily well-being using smartwatches. *IEEE Internet Computing* 24, 5 (2020), 10–17.
- [14] Tomaso Poggio, Hrushikesh Mhaskar, Lorenzo Rosasco, Brando Miranda, and Qianli Liao. 2017. Why and when can deep-but not shallow-networks avoid the curse of dimensionality: A review. *International Journal of Automation and Computing* 14 (12 2017), 1–17. <https://doi.org/10.1007/s11633-017-1054-2>
- [15] Attila Reiss, Ina Indlekofer, Philip Schmidt, and Kristof Van Laerhoven. 2019. Deep PPG: Large-scale heart rate estimation with convolutional neural networks. *Sensors* 19, 14 (2019), 3079.
- [16] Attila Reiss, Philip Schmidt, Ina Indlekofer, and Kristof Van Laerhoven. 2018. PPG-based Heart Rate Estimation with Time-Frequency Spectra: A Deep Learning Approach. 1283–1292. <https://doi.org/10.1145/3267305.3274176>
- [17] Giulio Ruffini, Ibanez-Soria David, Marta Castellano, Laura Dubreuil Vall, Aureli Soria-Frisch, Ron Postuma, Jean-François Gagnon, and Jacques Montplaisir. 2019. Deep Learning With EEG Spectrograms in Rapid Eye Movement Behavior Disorder. *Frontiers in Neurology* 10 (07 2019), 806. <https://doi.org/10.3389/fneur.2019.00806>
- [18] Seyed Salehizadeh, Duy Dao, Jeffrey Bolkhovsky, Chae Ho Cho, Yitzhak Mendelson, and Ki Chon. 2015. A Novel Time-Varying Spectral Filtering Algorithm for Reconstruction of Motion Artifact Corrupted Heart Rate Signals During Intense Physical Activities Using a Wearable Photoplethysmogram Sensor. *Sensors* 16 (12 2015), 10. <https://doi.org/10.3390/s16010010>
- [19] Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger, and Kristof Van Laerhoven. 2018. Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection (ICMI '18). Association for Computing Machinery, New York, NY, USA, 400–408. <https://doi.org/10.1145/3242969.3242985>
- [20] Tim Schäck, Christian Sledz, Michael Muma, and Abdelhak M. Zoubir. 2015. A new method for heart rate monitoring during physical exercise using photoplethysmographic signals. In *2015 23rd European Signal Processing Conference (EUSIPCO)*. 2666–2670. <https://doi.org/10.1109/EUSIPCO.2015.7362868>
- [21] Seok Bin Song, Jung Woo Nam, and Jin Heon Kim. 2021. NAS-PPG: PPG-Based Heart Rate Estimation Using Neural Architecture Search. *IEEE Sensors Journal* 21, 13 (2021), 14941–14949. <https://doi.org/10.1109/JSEN.2021.3073047>
- [22] Dimitris Spathis, Ignacio Perez-Pozuelo, Soren Brage, Nicholas J Wareham, and Cecilia Mascolo. 2021. Self-supervised transfer learning of physiological representations from free-living wearable data. In *Proceedings of the Conference on Health, Inference, and Learning*. 69–78.
- [23] Paul Van Gent, Haneen Farah, Nicole Van Nes, and Bart Van Arem. 2019. HeartPy: A novel heart rate algorithm for the analysis of noisy signals. *Transportation research part F: traffic psychology and behaviour* 66 (2019), 368–378.
- [24] Albert Vilamala and Lars K. Hansen. 2017. Deep Convolutional Neural Networks for Interpretable Analysis of EEG Sleep Stage Scoring. <https://doi.org/10.48550/ARXIV.1710.00633>
- [25] Menglian Zhou and Nandakumar Selvaraj. 2020. Heart Rate Monitoring using Sparse Spectral Curve Tracing. In *2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 5347–5352. <https://doi.org/10.1109/EMBC44109.2020.9175349>